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Modelling and design of ultra-high stable Fabry–Pérot cavity

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分析评估噪声源＋新技术

温度波动+布朗噪声

介电镜涂层和晶体镜涂层

ABSTRACT

Lasers having sub-Hz linewidth and long-term stability better than a few parts in 10^{16} is an unavoidable requisite to achieve unprecedented accuracy in the optical atomic clocks, intercomparing distributed atomic clocks and their networking, establishing coherent optical fibres, precision time & frequency transfer, production of optical qubits, and many more. For these wide ranges of applications, ultrastable Fabry–Pérot (FP) cavities, which act as the reference optical resonator, have got lots of attention. Designing such a FP cavity is a challenge due to the requirements of suppressing instabilities from all the thermo-mechanical noise sources at the highest degree. In this article, we report a comprehensive study to elucidate detailed design steps and associated numerical analysis procedures to build such a FP cavity. For this, we considered all sources of noises arising from: improper optical and opto-mechanical settings, ground vibrations transmitted to the cavity, deformations due to self-weight as it worsens the stability due to acoustic noise coupled to it, and temperature fluctuation. We present quantitative assessments of the instabilities for different shapes, geometries, and constituent materials of the cavities. In order to model the vibration-induced instability, we developed a novel forced-vibration based technique for accurate estimation of the instabilities, that reciprocates the experimental results reported in the literature, and the method is found to be superior compared to the other approaches, therefore estimating the overall cavity stability more accurately. We found in short term (< 10 s), the instability of the cavity is limited by temperature fluctuation, and in the long term (> 10 s), Brownian noise dominates the instability, therefore demands adequate precautions to reach the desired level of stability. Through systematic simulations, we have identified the best criteria for minimization of all these noises and obtain stabilities of 8.5×10^{-17} and 5.1×10^{-17} using dielectric and crystalline mirror coatings, respectively, which are similar to the best-reported results among its homologue cavities.

1. Introduction

Optical atomic clock enabled precision measurements to open up timekeeping with unprecedented accuracy [1], testing of the fundamental science [2–4] and the general theory of relativity [5–7], geodetic measurements [8], dark matter search [9] and gravitational waves [10], and many more. In the experimental realization of the optical clocks, ultrastable Fabry–Pèrot (FP) cavity is one of the key components to generate sub-Hz linewidth ultrastable reference lasers for exciting the clock transition. Therefore, an ultrastable FP cavity plays a vital role in building a state-of-the-art optical clock, which has reached an unprecedented level of accuracy [11,12]. The effort to achieve ultra-high stability of the FP cavity is practically limiting its optical path length change due to complex deformations resulting from various external thermo-mechanical loads. In order to achieve the desired stability on the order of a few parts in 10^{16} or better, such cavities must be meticulously designed and carefully isolated from various environmental perturbations such as seismic vibration, temperature, and pressure fluctuations. Designing an ultra-high stable cavity is therefore an engineering challenge that involves knowledge of mechanical, optical, and physical sciences.

Several approaches have been demonstrated in achieving ultrastable operation of these reference FP cavities. Long cavities [13–16] provide greater stability compared to the shorter ones [17–22], however, the former are dimensionally challenging for fabrication and operation. The short cavities can be made compact, therefore easy to control and maintain homogeneity of the environmental parameters over its dimensions. They are more favourable for making them transportable [23,24] and can be modified to a space-grade [25]. Thermal noise of the cavity

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materials is one of the primary sources to introduce instability to the FP cavity [26,27]. Frequency fluctuation due to the birefringence of highly reflective (> 99.998%) mirror coatings and cavity thermal noise can be reduced by increasing the length of the cavity, however, at the cost of increased vibrational instability. Since cooling reduces the Brownian noise of the materials, especially of mirror coatings [28], cryogenic operation of short cavities has also been demonstrated with ultimate performance so far [29–34]. The integrated system size for a cryogenic cavity is not small compared to a room temperature cavity due to the inclusion of the cryogenic chamber and it requires routine maintenance. The careful choice of materials for the spacer, mirror substrate and its coating is utterly important to reduce the impact of the various noise sources and efficient dissipation of the photon energy. For example, the choice of mirror substrate materials with low mechanical loss helps to reduce the thermal noise [35]. For that purpose, the crystalline coated mirrors on fused silica substrates (super mirrors) are more favourable for further reduction of thermal noise [36]. Thus, building an ultrastable FP cavity strongly depends on a meticulous selection of its materials, opting for the exact design parameters, choice of support structures, environmental isolation, etc., which are obtained by performing rigorous simulations and analysis before its fabrication.

Currently, laser frequency stabilization with respect to state-of-the-art room temperature cavities has achieved a frequency instability $\sim 10^{-16}$ in several laboratories across the world using different lengths, support systems, and shapes of the cavities. A few 100 mm long cavities were designed for their compact nature, transportability and for space applications in National Institute for Nuclear Physics (INFN), Italy (ADEV of 1×10^{-15}) [20], National Time Service Center (NTSC), China (ADEV of 7×10^{-15}) [24], Systèmes de Référence Temps-Espace (SYRTE), Paris (ADEV of 5.5×10^{-16}) [37], and the University of Birmingham, UK (ADEV of 7.9×10^{-16}) [38]. The Physics of the Interactions of Ions and Molecules (PIIM), France [39] developed a compact stable 150 mm vertically oriented conical cavity supported at its mid-palate having frequency instability of 9×10^{-15}. Longer stable room temperature cavities of spacer length between 250 to 300 mm were developed in East China Normal University (ECNU), China [16], Korea Research Institute of Standards and Science (KRISS), South Korea [40], National Research Council (NRC), Canada [41], among these the best-reported instability is $\sim 2 \times 10^{-16}$ by the ECNU. A longer cavity reduces the Brownian noise, which is one of the fundamental limiting factors for its instability. However, for a longer cavity, thermal insulation, firm mounting within a vacuum chamber, size of the pump, and overall dimension-cum-weight of the system becomes larger and unfriendly. A horizontal 485 mm long cavity is developed in the Physikalisch-Technische Bundesanstalt (PTB), Germany has achieved the lowest instability (ADEV) of 8×10^{-17} for room temperature operation [13].

Finite Element Analysis (FEA) is the most widely used numerical tool in design and analysis ranging from components to systems, some such example includes designing sensors [42], helmet [43], capsule robots [44] to space structures [45] and many more. In the quest to obtain the optimal design parameters for the FP cavity, FEA is therefore an obvious choice to investigate the merits and demerits of different configurations, evaluate designs and quantify the instabilities resulting from various sources [46,47]. A wide variety of designs in terms of the shape of the spacer [46,48], support structure [46,49], the orientation of the cavity [50], and material choices [47] have been investigated for efficient reduction of the instabilities. Horizontal and vertically mounted cylindrical, cubic, spherical, pyramidal and tetrahedral cavity shapes have been widely investigated [22,47]. Using FEA, Legero et al. [51] established that an annular ring optically contacted with the mirrors reduce their bending due to the mismatch in thermal expansion coefficients of the mirror and spacer materials, which was later implemented and demonstrated by various groups. Chen et al. [46] adapted static analysis to optimize the mounting structure for different cavities and concluded that the vibrational instability of vertical and horizontal (relative to the direction of gravity) cavities can be reduced with controlled machining and mounting tolerances. Wiens and Schiller [47] identified Poisson's ratio between 0.13 to 0.23 for the spacer material to nullify the influence of forces supporting the spacer. Further, they also reported that orientation along [110], [100] crystallographic directions exhibit low sensitivity to the manufacturing tolerances for crystalline spacer materials. Lee et al. [49] used FEA to study the vibration sensitivity of a cavity by varying the support position and area along with fixed support (fully constrained) and sliding support (only vertically constrained) conditions. They simulated vibration-insensitive support positions that are independent of the support area (namely, the Airy points) and compared those with the experiment. Therefore, it is evident that accurate quantitative analysis, using proper FEA and analytical techniques, for all sources causing instabilities to the cavity before its construction is necessary to ensure its ultrastable performance.

Here, we systematically present a comprehensive methodology to design an ultrastable FP cavity starting from accurately estimating the optimal spacer length, the careful choice of the constituent materials considering their relevant properties, and, finally numerical analysis to obtain instabilities caused by all sources of noises. In the process, we consider a variety of shapes, geometries, and mountings to obtain the optimum one(s) and discuss the results in detail in terms of their performance to hone a near room temperature state-of-the-art ultrastable FP cavity. In Section 2 optical calculations are presented to determine the length of the spacer (Section 2.1), followed by a material selection guideline for the FP cavity spacer and mirror to obtain the best thermo-mechanical stability (Section 2.2). The requisite methodologies to estimate various instabilities of the cavity from thermo-mechanical noises are formulated and results are presented together with novel findings in the Section 2.3. The results obtained from the analysis are discussed in the Section 3 and compared with the instabilities of the FP cavities reported across the world. Finally, the conclusions are drawn in Section 4.

2. Design and analysis

Optimum design parameters for an ultrastable FP cavity are obtained from rigorous optical and thermo-mechanical simulations. Optical calculations have been performed to determine the optimal length and acceptable machining tolerance of the spacer, radius of curvature of the mirrors for the interest of laser wavelength at 1550 nm, the surface quality of these elements followed by choice of materials for the spacer, annuli rings and mirrors. Finite Element Analysis (FEA) is performed to determine the shape of the cavity, optimal support points, mounting structure to maximize isolation of the acoustic noises, the optimum offset of the optical and spacer axes and minimizing its dynamicity in day-to-day characteristics. Further, the thermal shielding is designed using FEA so that the cavity is nearly insensitive to environmental temperature fluctuations, which otherwise dominate the instability.

2.1. Optical and opto-mechanical calculations

In this section, the necessary optical calculations are presented to choose the optimum length of the spacer, suppress optical powers distributed to the higher-order modes, and avoid their mutual overlapping. To understand this we start by presenting the stability phase diagram of dual mirror linear FP cavity, as shown in Fig. 1. Different geometric configurations (indicated within boxes of Fig. 1), with their parameters lying within the stable phase (coloured domain), allows the photons to resonate inside the cavity. Here, $\mathcal{G}_1 = \left(1 - \frac{L}{R_1}\right)$, $\mathcal{G}_2 = \left(1 - \frac{L}{R_2}\right)$, $\mathcal{G} = \sqrt{\mathcal{G}_1\mathcal{G}_2}$ and $0 \leq \mathcal{G} \leq 1$ for a stable cavity, where R_1 and R_2 define the radii of curvature of the mirrors and L is the length of the cavity. The ultrastable cavities have $\mathcal{G}$-values well within the stable region but they lie on the boundaries for the cavities that are used as sensors. In the latter case, the cavity is most sensitive to detect any

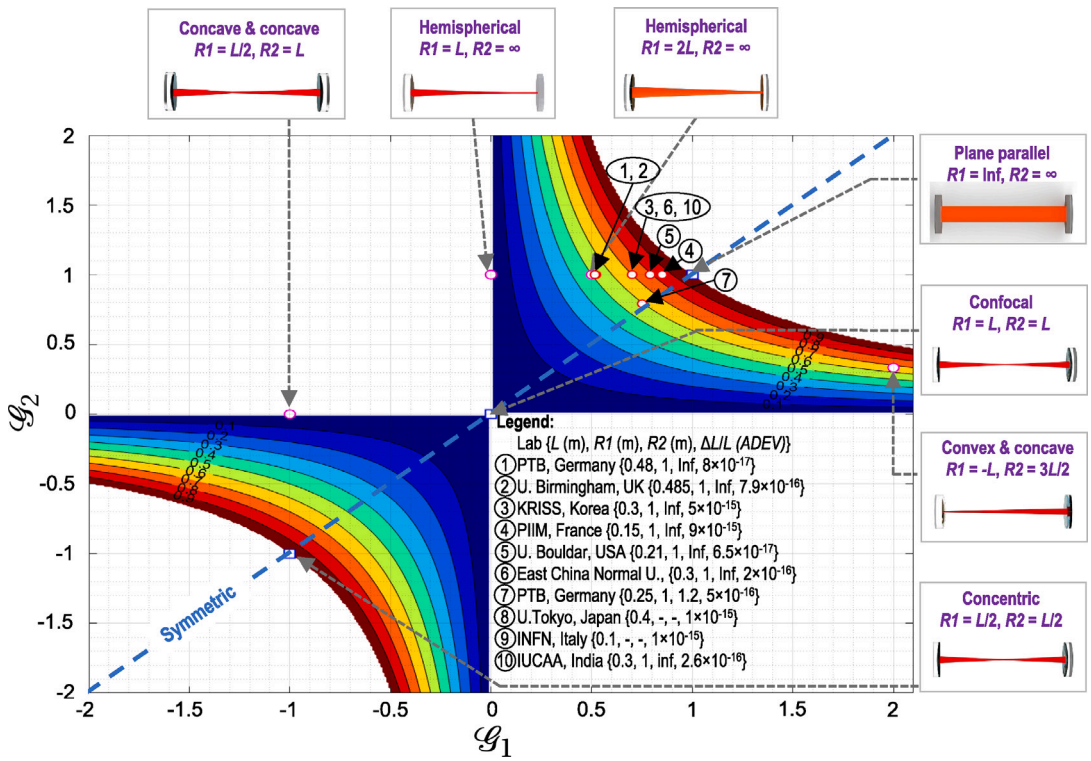

Fig. 1. Stability diagram of the Fabry–Pérot (FP) cavity, where $\mathcal{G}_1 = 1 - L/R_1$, $\mathcal{G}_2 = 1 - L/R_2$ and $\mathcal{G} = \mathcal{G}_1\mathcal{G}_2$ represent stability parameters (g-factors) of the cavity and $0 \le \mathcal{G} \le 1$ forms a stable cavity (coloured region). Different dual mirror cavity configurations are used either as optical references or as sensors shown in the boxes. Stabilities of some state-of-the-art reference FP cavities used in optical atomic clock experiments are indicated in this plot together with their locations, L, R_1, R_2, and reported ADEV are also mentioned.

Table 1
Geometry of cavities having $R_1 = 1$ m and $R_2 = \text{inf}$.

A:	$L = 189.8$ mm, cylindrical cross-section with 100 mm diameter
B:	$L = 189.8$ mm, cuboidal cross-section with 50 mm sides
C:	$L = 293.4$ mm, cylindrical cross-section with 100 mm diameter
D:	$L = 293.4$ mm, cuboidal cross-section with 50 mm sides
E:	$L = 485$ mm, cylindrical cross-section with 100 mm diameter
F:	$L = 485$ mm, cuboidal cross-section with 50 mm sides

perturbations causing a tiny change in earth's gravity, which brings them out of stability. Therefore, an ultrastable high finesse FP cavity is used for linewidth narrowing [52] and mode cleaning [53] of lasers. For our optical clock application, where the stability of a FP cavity provides short-term reference, a hemispherical cavity having $R_1 = 1$ m and $R_2 = \text{inf}$ is opted for this purpose. Higher reflectivity r of the cavity mirrors results in longer sustaining of the photons inside of the cavity, which is characterized by 'finesse' $F = \pi r/((1-r^2))$. In the case of a FP cavity stabilized laser, the length instability $\Delta L/L$ of the cavity directly influences the laser frequency instability as $\Delta\nu/\nu = -\Delta L/L$ [14]. Brownian noise of the constituent materials is the fundamental limiting factor for cavity instability, which is lower for a longer cavity, however, thermal instability increases with its length [51]. Manufacturing limitation, efficient thermal insulation, and firm mounting of a long cavity within an ultra-high vacuum chamber also need to be considered for deciding on a suitable cavity length. Considering all these factors, we performed simulations for the cavities mentioned in Table 1,

To start with the optical simulation, we estimate the suppression factor,

$$\kappa_{nm} \equiv \left[1 + \left(\frac{2F}{\pi}\right)^2 \sin^2(n+m)\frac{\zeta}{2}\right]^{-\frac{1}{2}}, \tag{1}$$

which indicates the suppression quality of the Higher Order Modes (HOMs) relative to the Fundamental Mode (FM) of the cavity [54]. The cavity modes are represented by the Hermite–Gaussian (HG_{nm}) polynomials with indices m and n, $m = n = 0$ is for the FM having Gaussian profile, and $\zeta = 2\cos^{-1}\left(\pm\sqrt{g_1 g_2}\right)$ is the Gouy phase shift. The frequency separation between any two consecutive modes that is the transverse mode separation frequency is given as [14],

$$\Delta\nu \equiv \frac{\zeta}{2\pi}\Delta\nu_{FSR} = \frac{1}{\pi}\cos^{-1}\left[\pm\sqrt{\left(1-\frac{L}{R_1}\right)\left(1-\frac{L}{R_2}\right)}\right]\Delta\nu_{FSR}, \tag{2}$$

where $\Delta\nu_{FSR} = c/2L$ is the free spectral range and c is the speed of light in a vacuum. To minimize HOM degeneracy supported by a cavity, one has to choose L such that the HOMs are not only co-resonant with FM but also have negligible powers than the detector's noise floor. We calculated optical powers in the HOMs, i.e., $(m + n = 200)$ to find out the optimal length of the cavity, as shown in Fig. 2, which are normalized to the corresponding FM's power. From this, we obtain optimized lengths of the spacer: 0.116 m, 0.1898 m, 0.2934 m, and 0.485 m, within an acceptable manufacturing tolerance of ±100 μm. We choose a cavity length of 293.4(1) mm, which corresponds to a FSR of 510.89(17) MHz, for our final design, which will be considered in the rest of this article. To avoid frequency degeneracy of residual HOMs with any of the FM, we calculate the respective mode pattern for a given cavity length, as shown in Fig. 3. HOMs from six preceding FMs are considered for this and we found the closest HOM is 2.1 MHz away from the FM, which is about 300 times higher than the linewidth of the cavity spectrum.

Upon optimization of the cavity length, the vibration shall perturb its ideal settings and cause HOM degeneracy. Misalignment of the coupled laser light and variation of the optimum coupling is caused by rigid rotation and micro-movements due to low-frequency seismic waves at wavelength $> L$ propagating to the cavity through its supports. A pictorial representation of such a misalignment is shown in Fig. 4. In particular, the transverse misalignment in the axial plane of the cavity (XY-plane) will dominate $\Delta\nu/\nu$ compared to its rotational movement

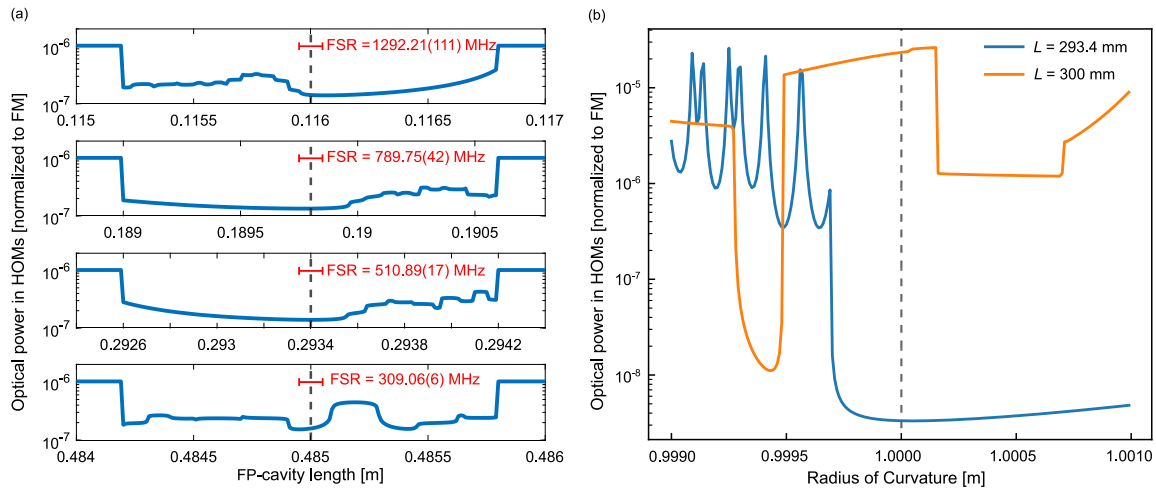

Fig. 2. Fabry–Pérot cavity length optimization: power in higher order modes (HOMs) normalized to that in the fundamental mode (FM). (a) Some of the optimized lengths are 116 mm, 189.8 mm, 293.4 mm, and 485 mm considering manufacturing length tolerance of ±100 μm as indicated by the error-bar (red), out of which we choose $L = 293.4$ mm for our cavity. (b) For a commercially available concave mirror having a standard radius of curvature of 1 m, power in HOMs is lower by several orders of magnitude for an optimized length $L = 293.4$ mm (blue) than that for a standard 300 mm length (red). This depicts the importance of length optimization to make the HOMs least prominent since otherwise, they are unavoidable even though undesired.

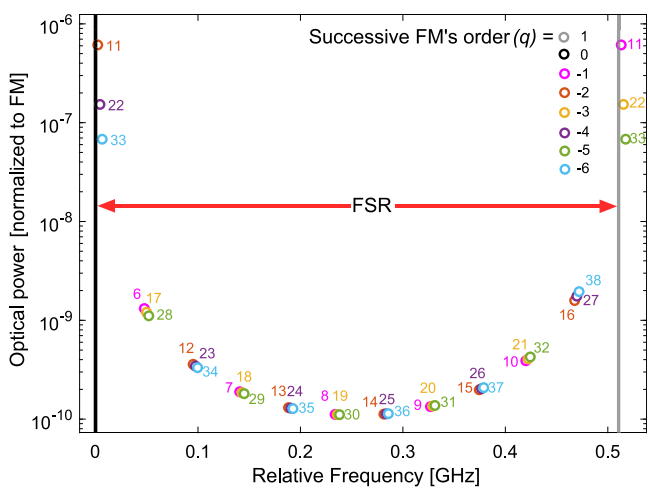

Fig. 3. Frequency spacing of the higher order modes HG_{nm}: discrete points show HOMs of successive FMs of order q, which are colour distinguished for different q, and the associated number in the legends give $(m+n)$. In our search up to $m+n = 200$ for an optimized cavity length the closest HOM is 2.1 MHz away from the FM, which is ∼ 300 times higher than the linewidth of the cavity spectrum.

even though L remains unchanged in both cases. If the axis of the cavity is displaced by $d_\circ$, then the electric field distribution in different modes due to the misaligned cavity is given by [55]

$$E \approx E_\circ \left[\left(1 - \left(\frac{d_\circ}{w_\circ}\right)^2 \left(1 - \frac{\sqrt{2}}{16}\right)\right) U_\circ - \frac{1}{2}\left(\frac{d_\circ}{w_\circ}\right) U_1 - \frac{\sqrt{2}}{16}\left(\frac{d_\circ}{w_\circ}\right)^2 U_2 \right], \tag{3}$$

where $E_\circ$ is amplitude, $w_\circ$ is the waist of an incoming Gaussian laser beam, $U_\circ$, U_1, U_2 are the fundamental and first two successive HOMs, respectively. In the case of an angular displacement θ, the electric field distribution is given by [55],

$$E \approx E_\circ \left[\left(1 - \left(\frac{\theta}{\theta_D}\right)^2 \left(1 - \frac{\sqrt{2}}{16}\right)\right) U_\circ - \frac{1}{2}\left(\frac{\theta}{\theta_D}\right) U_1 - \frac{\sqrt{2}}{16}\left(\frac{\theta}{\theta_D}\right)^2 U_2 \right], \tag{4}$$

where $\theta_D = \lambda/(\pi w_\circ)$ for the light at wavelength λ. The instabilities resulting due to $d_\circ$ and θ as given in Eq. (A.7) are determined (see Appendix A), and cumulatively considered in the simulated vibrational noise.

Variation in optical path length due to the fluctuations of inside pressure P and the radiation pressure also results in instabilities in the desired mode degeneracy. At a given temperature T, the pressure fluctuation ΔP is given by [56],

$$\frac{\Delta L}{L} = -\frac{dn}{dP}\Delta P, \tag{5}$$

follows $\frac{\Delta n}{n} \approx \frac{a}{2k_B T} 100 \Delta P$, where n is the refractive index of air, $a = \sqrt{k_B T/m}$ for m being the average mass of the gas molecule, and k_B is the Boltzmann constant. A pressure fluctuation of 1×10^{-11} mbar results to $\Delta L/L = 9\times10^{-18}$. Therefore, the vacuum vessel of the cavity has to be better than $\sim 10^{-7}$ mbar pressure and $\Delta P = 8\times10^{-11}$ mbar [14]. Instability due to radiation pressure on the mirror surfaces is given by [14]

$$\frac{\Delta L}{L} = \sqrt{\frac{2\left(1-\sigma_{fs}^2\right)}{\pi^{3/2} c Y_{fs}} h\nu P_{circ}}, \tag{6}$$

where $P_{circ} = \frac{P_{in}\, t\, r}{(1-r)^2}$ circulating optical power inside the cavity due to an incoming power P_{in}, t and r are transmissivity and reflectivity of the mirrors, h is the Planck constant, c is the speed of light in vacuum, σ_{fs} and Y_{fs} are Poisson's ratio and Young's modulus of the mirror substrate, respectively. In the case of a 1 μW input light, the corresponding circulating power is 1.5×10^{-2} W for the finesse of 1.5×10^5 results in $\Delta L/L = 8\times10^{-22}$. To minimize this effect, the mirror substrate has to have a higher mechanical quality factor, that corresponds to a lower loss angle, causing slower dissipation of energy impinging on it which is arising from P_{circ}.

2.2. Thermo-mechanical stability of the cavity

Following the identification of the spacer lengths, the choice of materials based on their properties is presented in this section. Mechanical and thermal stabilization of the cavity depends on self-weight, thermal fluctuation, seismic vibration, and dynamicity in earth's gravity, which are investigated for different geometry, shape, mounting structure, and materials to reduce instabilities caused by these thermo-mechanical loads. Relevant properties of a cavity, such as: effective

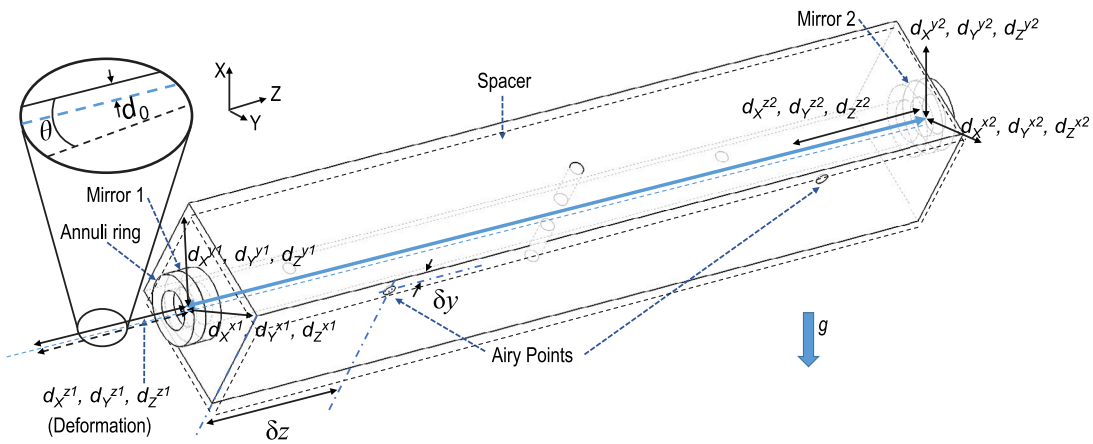

Fig. 4. Determination of the cavity support positions and its misalignment due to vibration, which are optimum at Airy points. Here, δy and δz represent the distances of Airy points measured from the edge of the cavity along y and z directions with respect to the direction of earth's gravity g for the two visible Airy points (1, and 2) in this view of the cavity those are distinguished by subscripts 1, and 2 in their respective symbols. The inset depicts a close-up view of the misalignment resulting in lateral ($d_\circ$) and angular (θ) displacements of the cavity.

Table 2
Effective coefficient of thermal expansion $\times 10^{-9}$/K, rate of change of cavity mode frequency with temperature ($\Delta\nu/\Delta T$) kHz/mK, and fractional frequency instability due to Brownian noise $\times 10^{-16}$ (at 1 Hz) for different combinations of spacer and mirror materials, respectively.

Mirror \ Spacer annuli	ULE	Fused Silica	Zerodur	Sapphire	Clearceram-Z
ULE	0.3 −0.06 0.94	487 −94.23 0.93	19.5 −3.77 1.1	5362 −1034.5 0.93	19.5 −3.77 1.1
Fused Silica	12.9 −2.49 0.45	500 −96.75 0.43	32.07 −6.2 0.73	5374 −1036.8 0.43	32.07 −6.21 0.74
Zerodur	0.8 −0.15 3.21	487.93 −94.41 3.21	20 −3.9 3.26	5362 −1034.5 3.21	20 −3.87 3.26
Sapphire	0.14 −26.7 0.21	625.69 −121.06 0.15	157.7 −30.52 0.61	5500 −1061 0.14	157.7 −30.52 0.62
Clearceram-Z	0.8 −0.15 3.2	487.93 −94.41 3.23	20 −3.9 3.29	5362 −1034.5 3.23	20 −3.9 3.29

Sitall [58] exhibits the same properties as Zerodur.

Coefficient of Thermal Expansion (CTE), temperature-dependent frequency drift $\Delta\nu/\Delta T$ that depends on the change in CTE with T, and Brownian noise depends on properties of the spacer, mirror, and annuli, which are listed in Table 2. Also, the aging of the materials and mechanical quality factor need to be considered for choosing the most suitable spacer-mirror-annuli material combinations. Dickmann et al. [57] demonstrated that using silicon as spacer material will result in frequency stability of 2×10^{-19}. The thermal noise of the reflective coating surface is the limiting factor for the total Brownian noise. The excellent stability using Silicon spacers comes with the cost of maintaining the cavity at either at 4 K or 124 K as those temperatures are zero crossing temperature of a monocrystalline silicon cavity. However, ULE is chosen mostly for its lower thermal expansion around its zero crossing temperature (5 to 35 °C), thus reducing frequency instability due to temperature fluctuation which is one of the major noise contributor ADEV at 1 s integration time. Thus, for a room temperature cavity, ULE spacer and FS mirror is found to be more suitable. Considering all these, we choose Ultra Low Expansion (ULE) glass and Fused Silica (FS) for the spacer-annuli and mirror materials, respectively, and a comprehensive comparison is given in Table 3. In the rest of the article, the results are obtained using ULE as the spacer-annuli and FS as the mirror substrate.

Table 3
Merits and demerits of different materials to use them as spacer, annuli and mirror.

Material combination for spacer, annuli - mirror	Remarks
ULE-ULE	ULE is not suitable as mirror due to its low quality factor and high Brownian noise
ULE-Zerodur/Sital	This combination gives high Brownian noise
ULE-FS	Suitable as the FS has high quality factor, relatively low Brownian noise and commercially available
ULE-Clearceram-Z	Clearceram-Z has higher Brownian noise and low quality factor
Zerodur/Sital-FS	Zerodur/Sital has faster aging

2.3. Finite Element Analysis (FEA)

This section elucidates FEA to investigate the thermo-mechanical deformations, which are performed using Ansys software [59]. Using FEA, the change in the optical path due to the deformation of the cavity (ΔL) is obtained and the resulting instabilities are calculated for different thermo-mechanical loads, which are presented in this section. Axially and radially symmetric cuboidal, cylindrical, and mushroom-shaped cavities [13–16] are the frequent choices, which also ease

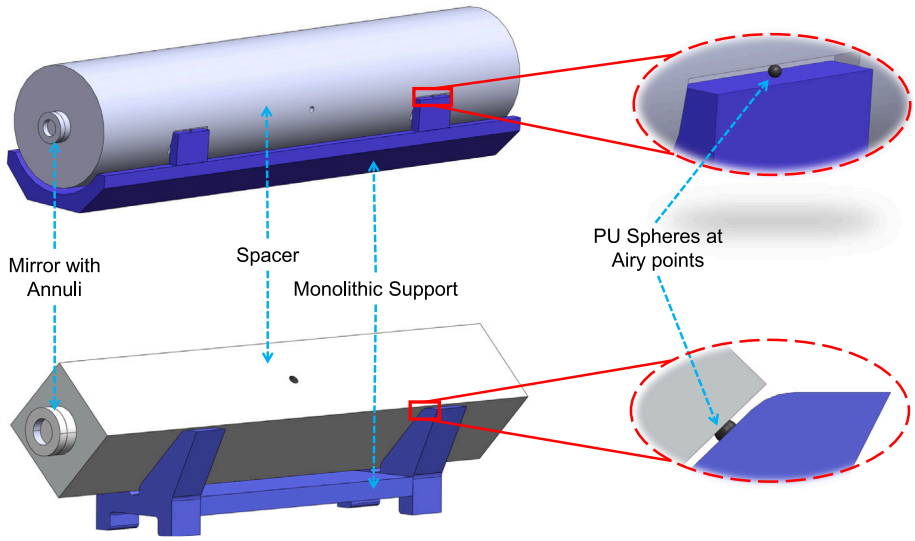

Fig. 5. Monolithic mounting structure (purple) for a cavity with cylindrical (top), and cuboidal (bottom) cross-sections of the spacers. The close-up view at the Airy point contacts shows Polyurethane (PU) balls (black) used to mount the cavity on top of their respective monolithic mounts.

Table 4
Material properties for FEA.

Parameters	Fused Silica	ULE	Polyurethane	Acrylic	Duraluminum
Density (kg/m^2)	2203	2210	1000	1180	2780
Young's Modulus (GPa)	73.1	67.7	0.1	2.69	73
Poisson's Ratio	0.17	0.17	0.5	0.4	0.33
Bulk Modulus (GPa)	36.92	34.2	4.5	4.27	78.5
Emissivity	0.8	0.745	–	0.94	0.03
Thermal Conductivity (W/mK)	1.38	1.31	–	0.19	130

the FEA. Here, cylindrical and cuboid cross-sectional spacers with monolithic support structures, as shown in Fig. 5, are reported even though other shapes were investigated and found not to be lucrative. Self balancing mount implemented by Hafner et al. [13] works well for heavier cavities. We found that for a 293.4 mm long cavity weighing around 2 kg, the natural frequencies of the mounting system have a considerable contribution to the total Power Spectral Density (PSD) displacement of the mirrors. The mounting system described by Keller et al. [60] shows promising results and further converting it into a monolithic support improved vibration insensitivity of the cavity system. In this work, the support structures are inspired by earlier works by Keller et al. [60] and Álvarez [14], however, are modified to make them monolithic. The cavity shall rest on four columns of monolithic support through intermediate polyurethane (PU) spheres of a diameter of 5 mm, which are placed on adequate grooves made on top of the facets of the pillars. The soft viscoelastic PU will facilitate in damping the vibration [61].

In FEA, duraluminum, ULE, FS, and acrylic are modelled as linear elastic materials, and their relevant properties are listed in Table 4, whereas, PU is modelled as a viscoelastic material using the Prony series representation [62] as listed in Table 5. Attachment of the mirrors to the annuli and the spacer is considered perfect bonding whereas other contact surfaces, e.g. between the supports and the spacer have a coefficient of friction of 0.5. The cavity model is meshed using 8-noded brick elements (Element 159 in Ansys). To efficiently mesh the model, we use the programme-controlled meshing and optimize it by using the in-built mesh optimization of Ansys, as an example shown for the annular, mirror, and spacer in Fig. 6(a). For any FEA, the mesh sensitivity analysis, as shown in Fig. 6(b–d) for the fractional instabilities due to self-weight, thermal fluctuation, and vibration, is necessary to avoid large simulation errors. Choosing a mesh where the results converge is important to get accurate results.

Table 5
Prony shear relaxation data for polyurethane [62].

Relative Moduli	1.85	4.04	15.04	43.28	107.45
Relaxation Time (s)	22.95	1.12	0.16	0.023	0.0023

2.3.1. Deformation due to self-weight

Deformation due to the self-weight of the cavity will cause deviation of its optical path from the manufactured length, as obtained from the optical & opto-mechanical optimization described in Section 2.1, which can be minimized by supporting the cavity at the Airy points [14,63] and proper positioning of the light coupled to it at an offset from its axis along the X-direction (Fig. 4) [60]. Non Airy point supports not only give larger initial deformation to the cavity but it enhances vibrational instability. As an example, Fig. 7(a) shows deformation of cavity D supported at the Airy points, as given in Table 1. Assuming the position of the Airy points are δy and δz along the Y and Z-directions (Fig. 4), respectively, their accurate determination is important. At the Airy points the cavity retains its initial position along the vertical direction (X-axis) and is relatively less deformed compared to the non-Airy points support, as shown in Fig. 7(b). The non-zero vertical deformation, on average, results from its initial gravitation sag due to its weight. On the other hand, each point along the longitudinal direction (Z-axis) of the cavity across its centre, which retains its position, is shifted on two opposite sides as a virtue of its vertical deformation. Such longitudinal deformation is minimum for Airy point supports, as shown in Fig. 7(c). A scanning of $\Delta L/L$ with respect to varying δy and δz is shown in Fig. 7(d) where the colour gradient indicates relative angular deformation between two mirror surfaces, which becomes minimum for the Airy points support and in that case the obtained $\Delta L/L = 3\times10^{-11}$ for the cavity-D. The change in $\Delta L/L$ due

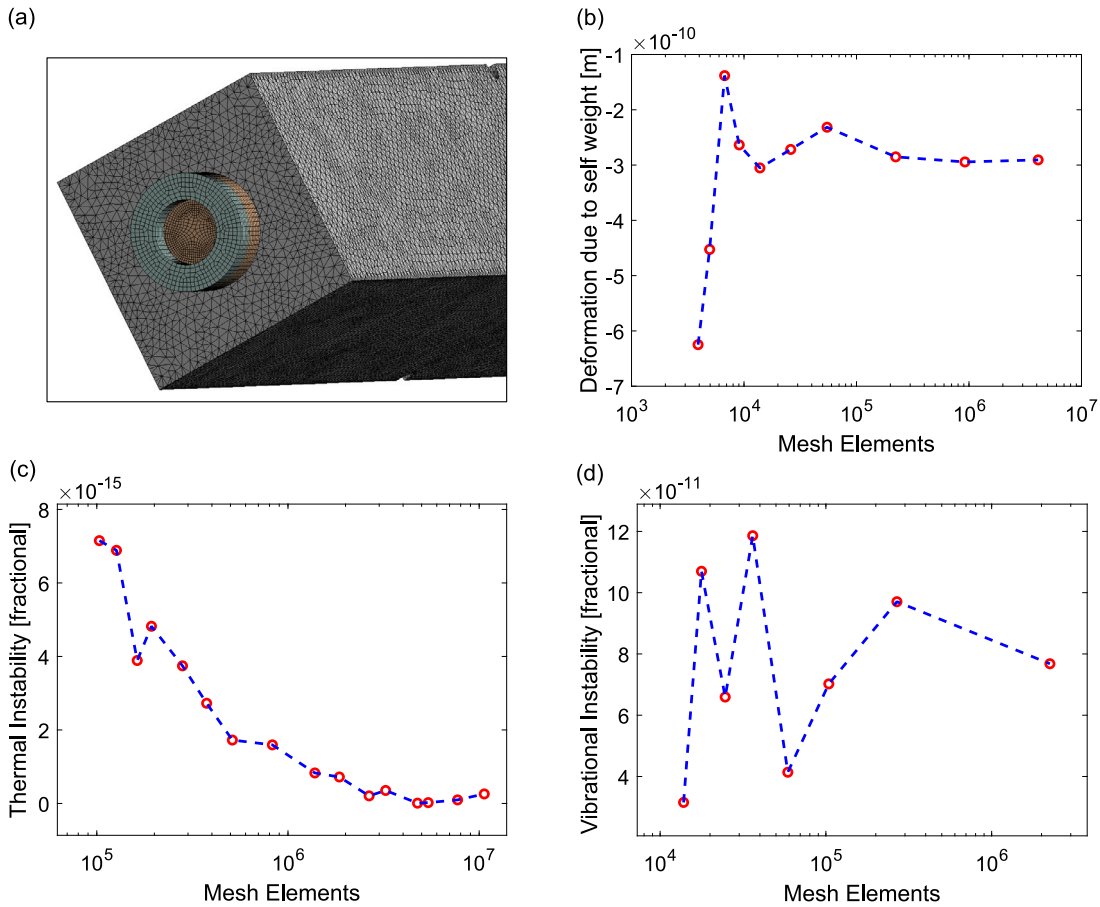

Fig. 6. (a) Representative finite element mesh for the cuboidal cavity, as an example, and mesh sensitivity study for correct estimation of the instabilities originating from (b) self-weight, (c) temperature fluctuation, and (d) vibration. hlThe predetermination of the correct mesh size is important to acquire accurate results at an optimum computation time.

to varying δy and δz that is a close-up view in the vicinity of the Airy point, as shown in Fig. 7(e), which suggests an acceptable tolerance to machine these points has to be within ±100 μm. The cavity length is also affected due to gravitational fluctuations Δg, which are caused due to low and high tides together with other factors [64]. For cavity-D that amounts to $\sim 4 \times 10^{-21}$ for a typical $\Delta g = 1.67 \times 10^{-10}$ m/s^2 [65].

Even though minimum deformation of the cavity length upon the Airy point support, yet there is a residual deviation of the optimized L as obtained using optical & opto-mechanical calculation. Further correction to it is possible by offsetting the mirror centres with respect to the spacer axis along the vertical direction (X-direction), which is necessary for minimizing the self-weight induced instability [60]. We performed the required simulation to obtain the correct value of that offset (δx) by varying the Airy points position (δy, δz), eventually, a 3-dimensional scan along all spatial directions, as shown in Fig. 8. It is evident from this figure that more than an order of magnitude improvement in $\Delta L/L$ is possible by placing the cavity axis 40 μm downwards to the spacer axis and that also fine-tunes the Airy point positions.

2.3.2. Temperature stability

Thermal expansion of constituent materials the FP cavity causes its instability due to change in optical path length. Minimization of such instability demands active mK level temperature stabilization and efficient thermal shielding to the extent of the characteristic heat flow time from the environment to the cavity. Therefore, the FP cavity is housed inside a three-layers heat shield, which is designed by using FEA. We opt for using a 10 mm thick duraluminium for making such heat shields, as shown in Fig. 9(a). Each of these three shields is maximally isolated from the other and their bottom layers are resting on the respective outer ones using three acrylic spheres of diameter 10 mm, which have thermal conductivity of 0.2 W/m/K. The plates have an adequate conical cutout with vent holes for the proper placement of these acrylic spheres and to avoid air trapping. Such an arrangement minimizes inter-shield surface contact and results in negligible thermal conduction. The entire cavity system inclusion of the thermal shield is housed inside a vacuum chamber that nearly stops the convection, which is thus ignored. The radiation plays the dominating role in the heat transfer, which is also minimized by polishing the heat shield surfaces that reduce their emissivity coefficient to 0.03 [66]. Fig. 9(b) shows the computed characteristic time of 6.53 days that is resulting in 15 days for thermal stabilization, which is similar to the other reported results [13].

In addition to proper thermal shielding, active temperature stabilization of the cavity system is essential. The "effective" CTE is determined by the thermal expansion properties of spacer-mirror-annuli, which depend on their individual CTEs, manufacturing process, and geometry. The effective CTE of a cavity system becomes nearly zero at a certain temperature $T_\circ$, namely the zero crossing temperature, as shown in Fig. 10. In this case, it is calculated using FEA, however, needs to be determined in practice as it varies due to batch-to-batch production of the materials. Typical CTEs of ULE and FS follow the temperature dependence as,

$$\begin{aligned} \alpha_{ULE} &= (T - T_\circ^{ULE})\, 2.4 \times 10^{-9} \\ \alpha_{FS} &= (T - 294.15)\, 2.2 \times 10^{-9} + 500 \times 10^{-9}, \end{aligned} \tag{7}$$

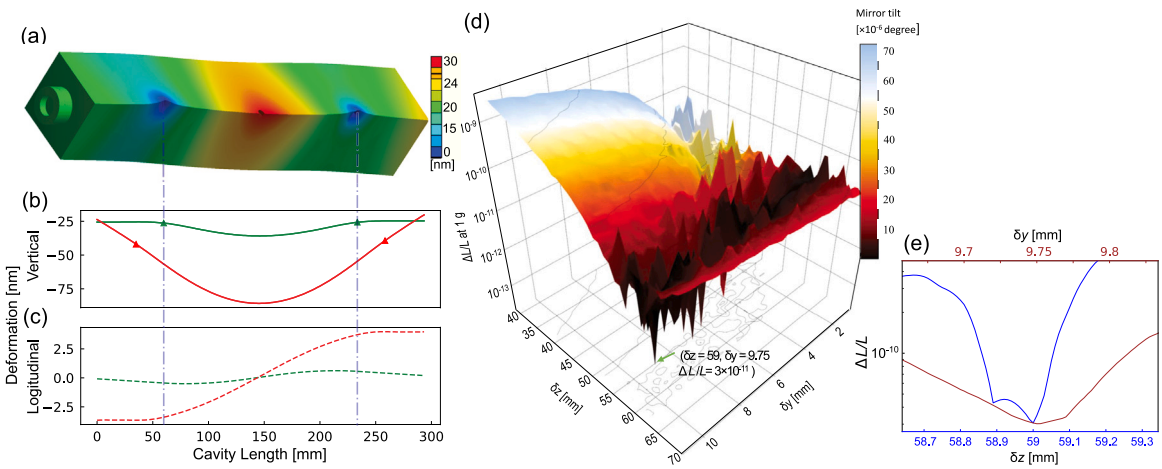

Fig. 7. (a) Deformation due to self-weight of the cavity-D (see Table 1 for the details) when supported at the Airy points. (b) Spatial dependence of the vertical (X-direction) and (c) longitudinal (Z-direction) deformations along the axis of the cavity-D (that is the light propagation direction), when supported at the Airy points (green) and non-Airy points (red) those are arbitrarily chosen 20 mm away from the Airy points. (d) A 2-dimensional scanning of the support positions (δz, δy) showing relative deformation of the cavity length $\Delta L/L$ and the minimum of that occurs at the Airy points. Here, the colour represents relative angular deformation between two mirror surfaces which shifts the optical path length from its desired value. (e) Variations of $\Delta L/L$, in the vicinity of the Airy points, for varying δz (blue) and δy (brown) when the respective other directions are fixed at the Airy points.

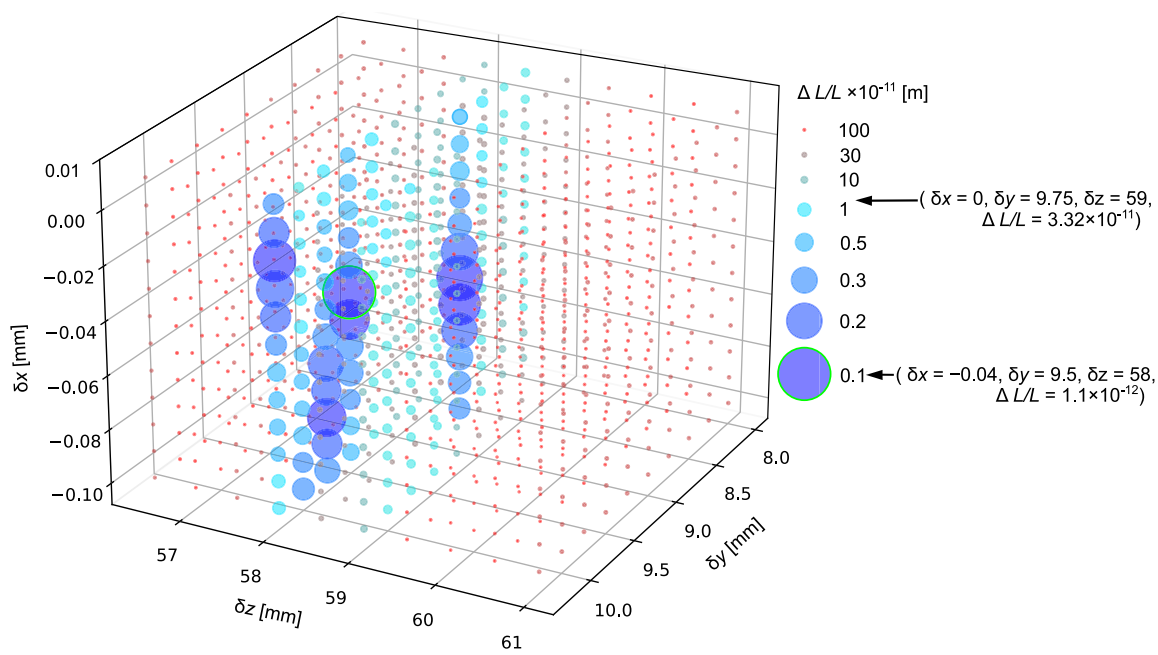

Fig. 8. Upon Airy point support of the cavity-D (as shown in Fig. 7), further improvement of stability by offsetting mirror centres δx with respect to the spacer axis along the vertical direction (X-direction). The bigger disc size in this plot represents smaller $\Delta L/L$ in addition to colour mapping of their values and the least of that as indicated by the green circle, is the optimum offsetting of the mirror centres with respect to the cavity axis.

where $T_{\circ}^{ULE}$ is zero crossing temperature of ULE and modified to $T_{\circ}$ in a spacer-mirror-annuli-based cavity system. The steep change of the CTE across $T_{\circ}$ demands active temperature stabilization of the entire cavity system within a few mK range across it. For example, ±3 mK across $T_{\circ}$ gives rise to an order of magnitude change in frequency instability, as shown in the inset of Fig. 10. To overcome this, we plan to use a precision temperature controlling-cum-stabilization, as described in Ref. [67]. The deformation of the inner surface of the mirror due to a 1 mK rise from its effective zero crossing temperature ($T_{\circ}$) is shown in Fig. 11.

2.3.3. Seismic vibration

In this section, the vibration induced instability of the FP cavity is investigated in detail. Stabilization against vibration is another important factor to acquire the ultrastable behaviour of a FP cavity, like many other high precision equipments [68–70]. However, in this case, stability cannot be achieved by vibration isolation only but depends on mounting structure, precise determination of the point to hold the cavities, and materials to be used at the point of contact. This requires a specially designed mounting structure, placing the entire system on a load specific passive/ active mechanical vibration isolator that is capable to suppress ideally 0 to ∼100 Hz acoustic noise and keeping it

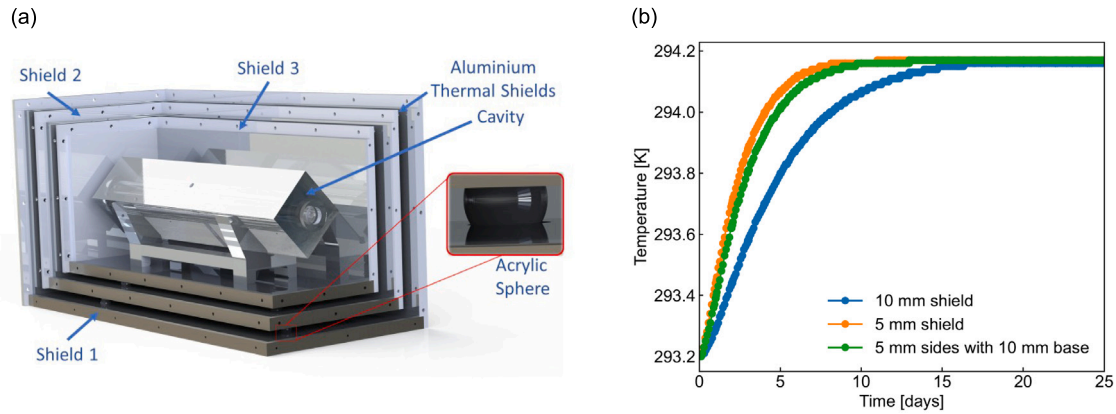

Fig. 9. (a) FP cavity is housed inside three layers of thermal shielding, those are three independent enclosures made of duraluminum with their surfaces mirror-like polished and mutually separated by using acrylic balls at their bottom surfaces, as shown within the close-up view. (b) Rise of the temperature inside of the inner shield (shield 3) for 1 K change in the ambient temperature (outside the shield 1) for different thicknesses of the enclosures: 10 mm (blue), 5 mm (orange), and 5 mm all sides on top of a 10 mm base (green).

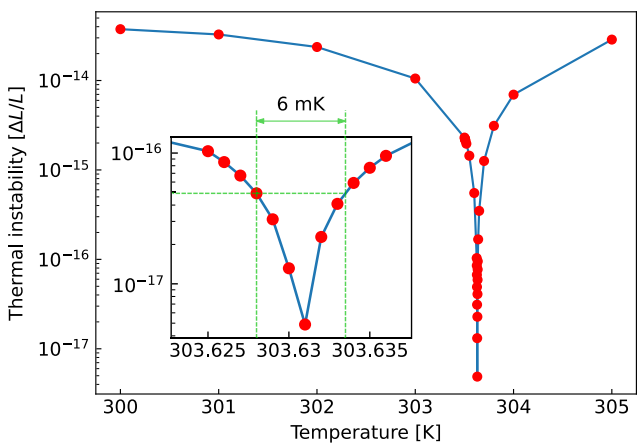

Fig. 10. Fractional length instability $\Delta L/L$ due to thermal expansion of a fixed cavity length L across the effective coefficient of thermal expansion (CTE) of the entire spacer-mirror-annuli cavity system. The zero crossing temperature of the effective CTE is the temperature corresponding to the minimum thermal expansion. The inset shows a close-up view around the zero crossing temperature to highlight the fact that $\Delta L/L$ is increased by one order of magnitude due to ± 3 mK change in temperature from there, therefore critical requisite of active thermal control of the system.

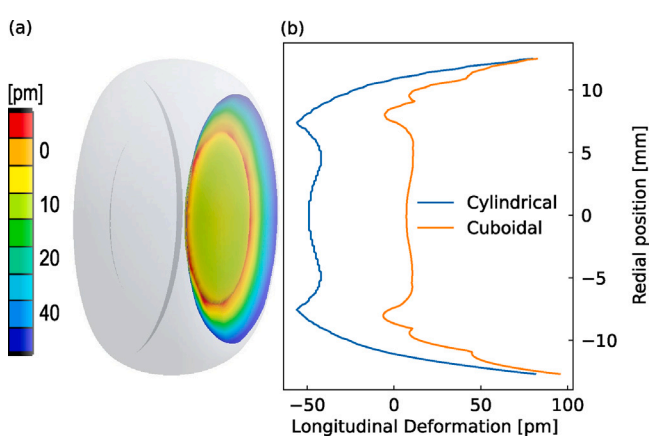

Fig. 11. (a) Simulated thermal deformation of the mirror surface for 1 mK rise from its zero crossing temperature, and (b) corresponding longitudinal deformation of the mirror inner surface along the diameter for cavity-C (cyan) and D (orange), having the cylindrical and cuboidal cross-section of their spacers, respectively. Therefore a cuboidal cavity is more suitable (Table 1).

within an acoustically clean enclosure. FEA is performed to evaluate the effectiveness of different mounting structures, as shown in Fig. 5, and their constituent material properties are given in Table 4. We consider a typical ground vibration data of our Laboratory, as shown in Fig. 12(a), with and without an active mechanical vibration isolator, which is used as the reference to obtain the effectiveness of our designed mountings.

We studied different approaches for estimating the vibration-induced instability and validated one method against the reported experimental data in Ref. [14]. The commonly reported method obtains the acceleration sensitivity either against the gravity (vertical direction) [14] or along all three mutually perpendicular directions and considers the largest one as the vibrational sensitivity of the cavity [71], which we named Method-1 and 2, respectively. For simulations, an acceleration same as g is applied on the FP cavity maintaining the boundary condition: the Airy points are fixed supports. At these points, circles of 1 mm diameter on the spacer surface are considered to ensure the forces resulting due to g are normal. As a result, the normal component of the acceleration stabilizes against gravity, however, the longitudinal components result in the instability of the cavity. The centre-to-centre relative deformations between the mirrors are extracted to estimate the acceleration sensitivity of the cavity. In these two methods, modal or natural frequency analysis is not performed. The random vibration analysis, that is Method-3, uses Power Spectral Density (PSD) as an input along the surface of contact to the vibrational isolation platform (that is the fixed support: the boundary condition in this method) after it is suppressed by the same (Fig. 12(a)). FEA for Method-3 first goes through the modal analysis which determines the dominant 99 natural frequencies of the system that consist of the cavity, mounts, thermal shields, and vacuum enclosure within 0–2000 Hz bandwidth. The obtained vibration modes are considered for calculating the displacement at the centre of the mirrors and also at the Airy points as shown in Fig. 12(a). We performed mode number and mesh size analysis, however, it is computationally taxing with increasing mode numbers and finer mesh. In this method, using the random vibration module, displacement of the mirrors cannot be extracted explicitly.

In the forced vibration method (Method-4), modal analysis is used, however, unlike Method-3 here we provide acceleration to the cavity in a sinusoidal pattern at certain discrete frequencies e.g. 0.1 Hz, 1 Hz, 2 Hz, 10 Hz, 100 Hz, and 1000 Hz in all three directions. The time-dependent displacement is then determined using the transient structural module of Ansys. This allows us to determine if both the mirrors are vibrating in or out-of-phase, which makes it more advantageous than Method-3. In-phase vibrations, which are dominant < 10 Hz, cause instability due to misalignment whereas out-of-phase vibration results in a change in ΔL. To establish the state of confidence

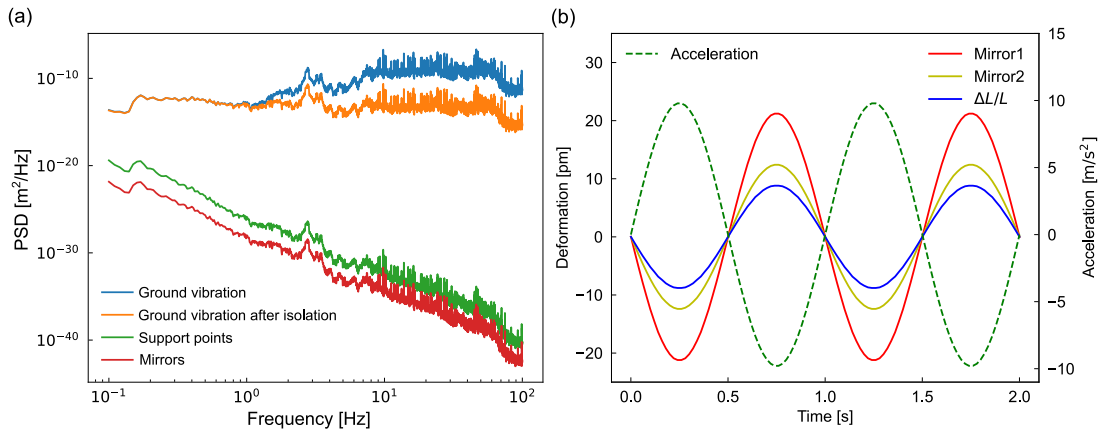

Fig. 12. (a) Power spectral Density (PSD) of vibrations at different locations: ground (blue), on top of a commercial actively controlled vibration isolation platform sitting on top of a granite block available in the lab (orange), at the point of support on the monolithic mounting structure while the system is placed on top of the vibration isolation platform (green), and at the cavity mirrors when the entire system is housed as shown in the graphical abstract (red). (b) The vibration results in the vectorial deformation of the centre of the mirrors when subjected to a g (peak-to-peak) sinusoidal acceleration (body force) at 1 Hz along the vertical axis to the cavity-D.

Table 6
Instability of the cavity described in Ref. [14] due to vibration calculated using different methods, which considers g acceleration acting on it.

Method-1	Method-2	Method-3	Method-4	Experimental [14]
2.5×10^{-11} (this work) 7×10^{-12} [14]	2×10^{-10}	4.55×10^{-11}	5.84×10^{-10}	5.9×10^{-10}

in proposed methods (1 to 4), we used the simulated and experimental results reported in Ref. [14] as a test bed and compared those with our simulations obtained via four different methods, as listed in Table 6.

Method-1 was used in Ref. [14] and also carried out by us independently. The difference in the values may be attributed to mesh sensitivity, particular boundary conditions at the viton support and material properties of the viton. Further, the values reported in Ref. [14] are also away from experimental results. This, in fact, motivated us to seek alternative models to quantify the vibration-induced instability. From Table 6 it is evident that Method-4 reciprocates the experimental result as it considers frequency-dependent acceleration sensitivity of the cavity in all three directions providing a better real-life representation. The deformation at the centre of the mirrors due to 1 Hz sinusoidal acceleration at g on cavity-D is shown in Fig. 12(b). Using Method-4 (a detailed description of this method is given in the supplementary document of this article) we obtain the resultant vibrational instability 5.8×10^{-18} for cavity-D. Method 4, based on the forced vibration analysis, proves to be advantageous and accurately captures the instability as it allows to account for in-phase and out-of-phase displacements of the mirrors as well as vibration in three directions, which is more realistic.

3. Discussion of results

Designing an ultrastable FP cavity starts with obtaining the optimal L for a fixed set of mirror geometries and configuration (Fig. 1) to reduce optical powers by about seven orders of magnitude in the HOMs relative to that in the FM. Length optimization also includes frequency overlapping of any HOMs with the resonant FM to avoid fricking of the Pound–Drever–Hall (PDH) lock from the desired FM to its co-resonant HOM(s). The necessary calculations for all these also gauge the requisite machining tolerances, relative parallelism of the spacer facets as well as surface flatness of spacer facets and annuli. We recommend tolerance on L to be ±100 μm to fulfil the optimal optical designs, the surface roughness of the spacer and annuli has to be about 20 nm for the optical contacting. We exercised different cavity geometry, as listed in Table 1, to find the most suitable one, which is compared in Fig. 13.

The next step is the proper choice of materials for the construction of spacer, annuli, and mirror substrate considering the properties of the combined system that should lead to the desired stability. For this, we considered three parameters: (i) effective CTE of the entire spacer-annuli-mirror system, (ii) $\Delta\nu/\Delta T$ resulting from T dependence on the effective CTE across its T_o, and (iii) Brownian noise of the cavity system which is dominated by the mirror coating, which is listed in Table 2: the material selection guide for different combinations. For practical purposes, one has to consider material availability, the possibility of machining and polishing, aging, and quality control in batch-to-batch production as they are proprietary items. From Table 2, it appears the entire system made of ULE is superior, however, FS having about two orders of magnitude higher mechanical quality factor than the ULE suits it best for the mirror substrate. Radiation pressure on the highly reflective mirror coatings does not play any significant role, unlike for the gravitational wave detecting interferometers, due to the low input of the laser power in our application. Rather, mirror coatings having low Brownian noise, e.g. crystalline coatings give about factor two better long-term stability as it is estimated for cavity-D.

Upon optimization of the optical and opto-mechanical parameters, instabilities of the cavity due to thermo-mechanical loads are studied. The change in optical path length due to deformation of the cavity is determined considering three major aspects: deformation due to self-weight, temperature fluctuation, and ground vibration coupled to it. We show that Airy points supported cavity results in the least deformation due to self-weight and also minimize angular deformation of the mirrors relative to each other (Fig. 7). Yet instability due to the change in optical path length resulting from residual deformation of the cavity along its axis can further be corrected by offsetting the mirror centres along the gravity direction with respect to the spacer axis (Fig. 8). We find self-weight deformation reduces for a cuboidal shaped longer cavity (Fig. 13). Temperature controlling within a few mK from the desired set point T_o is the most important to limit cavity stability within a few parts in 10^{16}. Short time-scale (fluctuation rate > 1 mHz) temperature fluctuations are eliminated by efficient thermal shielding, which in our system obtains a $1/e$ time period of 6.53 days. However, temperature drifts over a longer period (fluctuation rate < 1 mHz) are eliminated only using active stabilization. We considered temperature stabilization within ±5 mK, which obtains ADEV of 2.42×10^{-16} at 1 s for the cavity-D and similar for all other geometries (Fig. 13) except minute variations due to different T dependence of their respective effective CTE across T_o. Acoustic vibrations coupled to the cavity cause its rotation across all three axes, translate as a whole along all the directions as well as deform its shape. We find the forced vibration approach (method-4), is superior among all other approaches as it

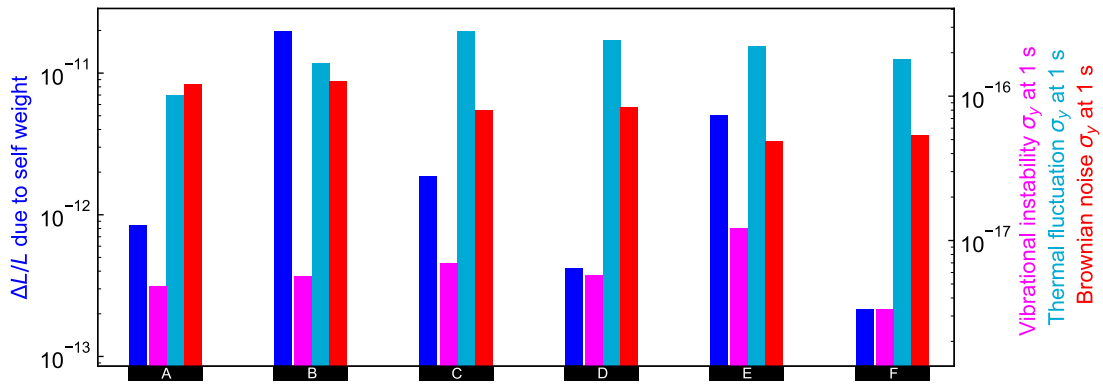

Fig. 13. Instabilities σ_y resulting from fractional length change ($\Delta L/L$) due to self-weight (blue), vibration (magenta), temperature fluctuation (cyan), and Brownian (red) for all types of cavities studied in this work (Table 1). This gives an at-a-glance comparison of their qualities from different perspectives.

Table 7
Instability budget from different noise sources, as an example for the cavity-D (refer Table 1).

Noise source	Parameter estimation	ADEV[a] @ 1 sec ($\times 10^{-17}$)
Vibration	Estimated considering both misalignment and deformation of the cavity using Eq. (A.10).	0.576(4)
Temperature fluctuations	Considers temperature stabilization within ±5 mK across $T_\circ$	24.2(2)
Brownian	$S_{total} = S_{spa} + S_{sub1} + S_{sub2} + 2S_{coat1,2}$ m^2/Hz. Spacer: $S_{spa} = \frac{4k_BTL}{2\pi f Q_{spa} Y_{spa} A}$ Mechanical Quality factor of spacer $Q_{spa} = 62500(625)$, Young's Modulus of spacer $Y_{spa} = 67.7(7)$GPa, Spacer cross-section area A. Substrate of both mirrors: $S_{sub1,2} = \frac{4k_BTL(1-\sigma^2)}{2\pi f Q_{sub}\sqrt{\pi} Y_{sub}\,\omega_{M1,2}}$ Where, Poisson ratio, Young's modulus, Mechanical quality factor of mirror substrate are $\sigma = 0.170(2)$, $Q_{sub} = 1.00(1)\times 10^6$ and $Y_{sub} = 73.1(6)$GPa, respectively. Beam radius on both the mirrors are ω_{M1} and ω_{M2}. Coatings: $S_{coat1,2} = \frac{4k_BT2(1+\sigma)(1-2\sigma)d_{coat}}{2\pi^2 f Q_{coat} Y_{sub}\omega_{M1,2}^2}$ Thickness d_{coat}, mechanical quality factors Q_{coat} of the dielectric and crystalline coating are 5.24μm, $2.50(3)\times 10^3$ and 1×10^5, respectively.	Dielectric: 8.430(3) Crystalline: 5.043(1)
Electronic hardware	$S_E = \sqrt{\frac{(S_{laser}+S_{act})^2+(KS_{servo})^2+(KGS_{disc})^2}{(1+KGD_v)^2}}$ Laser frequency noise ASD $S_{laser} = 16.57(17)$ Hz/$\sqrt{\text{Hz}}$ Actuator frequency noise ASD $S_{act} = 24.23(24)$ Hz/$\sqrt{\text{Hz}}$ Voltage to frequency conversion parameter $K = 2.00(2)\times 10^9$ Hz/V Servo voltage noise ASD $S_{servo} = 1.01(1)\times 10^{-6}$ V/$\sqrt{\text{Hz}}$ Servo gain $G = 10.0(1)$ V/V Frequency discrimination noise ASD $S_{disc} = 1.13(1)\times 10^{-9}$ V/$\sqrt{\text{Hz}}$ Frequency discriminator slope $D_v = 8.15(42)\times 10^{-5}$ V/Hz Here, KS_{servo} dominates.	3.61(4)
Residual Amplitude Modulation (RAM)	Electro-optic modulator with an angled facet is considered for the PDH setup to reduce RAM-induced frequency instability.	0.0171(3)
Radiation pressure	$S_{rp} = 2hfP_{circ}\left[\frac{2(1-\sigma^2)F}{\pi^{3/2}cY_{fs}\omega_M}\right]^2$ Circulating power in cavity $P_{circ} = 11.45(13)$ W, Input power is 1 mW, Cavity Finesse $F = 1.5032000(3)\times 10^5$.	0.001913(1)
Variation in earths gravity	Typical gravity fluctuation of 1.67×10^{-10}	0.0003
Total	Dielectric mirror coatings Crystalline mirror coatings	25.9(3) 24.9(3)

[a]ADEV for fractional frequency instability.

captures all of the above-mentioned vibration induced perturbations, further, the obtained results can be decoupled to extract instabilities of the cavity caused by misalignment of the light coupled to it and due to its physical deformation. Therefore, inevitably this method can reproduce the experimental results (Table 6), and using this we obtain ADEV of 5.8×10^{-18} (ADEV) at 1 s for cavity-D. The Brownian noise, in a longer time scale that determines the stability of the cavity in case of no aging, is lower for a longer cavity (Fig. 13). With the use of dielectric mirrors cavity-E, F obtains Brownian noise limited stability of 5×10^{-17}, which is 8.5×10^{-17} for the cavity-D and that is further reduced to 5.1×10^{-17} with the use of crystalline mirrors.

Simulated frequency responses of different noise sources resulting in the cavity instability, individually as well as their sum, are depicted in Fig. 14(a) for cavity-D. The frequency dependence of both the lab vibration (Fig. 12(a)) as well as the cavity's response to it for an optimized design, gives overall frequency-dependent instabilities induced by vibration and variation of gravity. The frequency dependency of noises from Brownian and the electronic hardware associated with the PDH lock is given in Table 7. Electro-optic modulator (EOM) with an angled facet is considered for the PDH loop (a detailed description is given in the supplementary document of this article), will generate RAM and the associated ADEV noise is depicted in Fig. 14(b) The instabilities caused by temperature fluctuation and radiation pressure are frequency independent. The ADEVs corresponding to all these instabilities are calculated following the technique mentioned in Ref. [72] and are given in Fig. 14(b) together with their quadrature sum. The instability budget for cavity-D is given in Table 7 and the resultant ADEVs with the use of dielectric and crystalline mirrors are shown in Fig. 14(c) for clearer visualization and comparison. Our theoretical result is compared with the reported stabilities of different ultrastable FP cavities, operating near room temperature and cryogenic, around the world and depicted in Fig. 15.

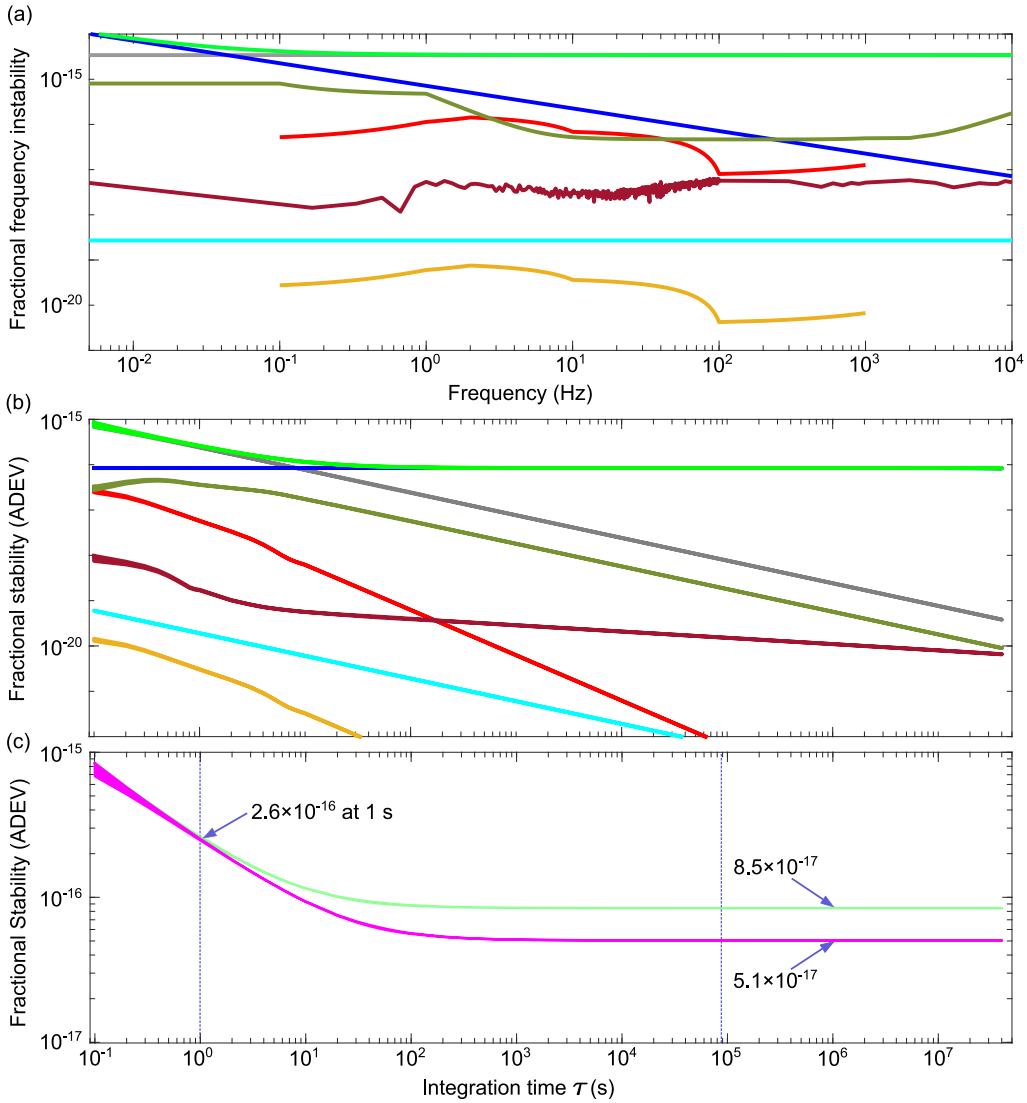

Fig. 14. (a) Frequency response of different sources of instabilities and (b) corresponding ADEV for temperature fluctuation (grey), Brownian (blue), electronic hardware (olive green), vibration (red), Residual Amplitude Modulation (RAM) (brown), radiation pressure (cyan), gravity g variation (yellow) and their quadrature sum (green) for the 293.5 mm cuboidal cavity (cavity-D). (c) Resultant ADEV considering dielectric (green) and crystalline (magenta) mirror coatings for cavity-D, where crystalline mirrors are slightly better due to the lower Brownian noise of their mirror coatings. The vertical lines (dotted blue) depict 1 s and 1 day.

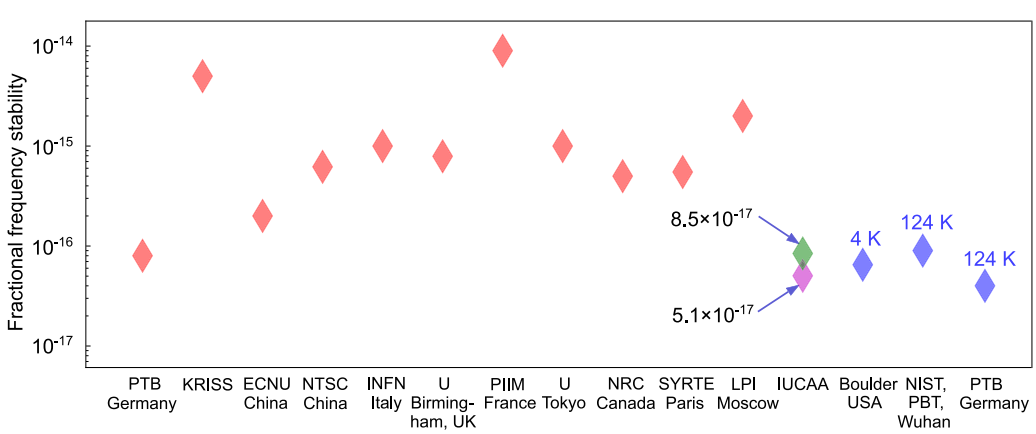

Fig. 15. Reported stabilities of FP cavities: near room temperature cavities (red), cryogenic cavities (blue) by different groups across the globe, and our results considering dielectric (green) and crystalline (magenta) mirror coatings.

4. Conclusion

We describe methodological ways of designing a state-of-the-art ultrastable Fabry–Pérot (FP) cavity through optical calculations and finite element analysis considering all possible noise sources that affect the cavity stability. The optimal spacer lengths are determined from the optical calculations that ensure negligible energies in the HOMs minimize instability due to the unavoidable mode degeneracy. Accurate optical calculations are also necessary to avoid overlapping of HOMs with successive FM which is the desired lock reference since otherwise, the locking electronics do not distinguish them. In the next step, the consequences of the choice of constituent materials, and shapes of the cavity-cum-mounting structures are quantitatively evaluated from the simulations. In course of that, we considered different geometries and shapes of the cavity as well as for the support structures. Further, effective thermal shielding and mechanically induced instabilities due to the deformation from self-weight, thermal fluctuation, dynamicity in earth's gravity, and, seismic vibration are simulated. The numerical results are compared with literature where available. Therefore, we give a detailed description to acquire an optimized solution for building an ultrastable cavity, which shall provide comprehensive guidance for building such cavities.

We have identified the correct approach to model the vibration-induced instability appropriately and verified results obtained through our model with the reported experimental results. This allows accurate simulation of the vibrational instability thereby helping to hone the ultrastable characteristics of the cavity. We also find that a few mK level active temperature stabilization of the entire system, even if it is enclosed inside of an efficient thermal insulation, is a necessity to achieve ultrastable features since this is dominant over other sources of instabilities. Radiation pressure and gravity fluctuations are the least important for that manner and all instability resulting from all possible noise sources are depicted in a single plot as well as tabulated for at-a-glance comparison.

Upon comparing a wide variety of cavity shapes, geometries, and mountings our opted design, that is cavity D (293.4 mm long and cuboidal spacer crosssection), with optimized instabilities from all mentioned sources has achieved a stability of $25.9(3) \times 10^{-17}$ and $24.9(3) \times 10^{-17}$ at 1 s with the use of dielectric and crystalline mirror coatings, respectively. They finally reach to maximum stability of 8.5×10^{-17} and 5.1×10^{-17}, which are dominated by the Brownian noise of the mirror coatings. These obtained results are comparable to the best room temperature cavity [13] and better than many others reported by various groups worldwide and, as expected, slightly away from the state-of-the-art cryogenic silicon cavity that has reached 4×10^{-17} [33]. Upon consideration of all minute details in a methodological way, we report the design of an ultrastable FP cavity that can be used for building a narrow linewidth ultrastable laser, which is an unavoidable requisite for optical atomic clock, high precision laser spectroscopy, and many more.

CRediT authorship contribution statement

Sankalpa Banerjee: Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Visualization. **Stanley Johnson:** Methodology, Formal analysis, Writing – original draft. **Yutiben Vaghasia:** Methodology, Formal analysis, Writing – review & editing. **Kanik Palodhi:** Writing – review & editing. **Sandip Haldar:** Conceptualization, Methodology, Formal Analysis, Writing – review & editing, Supervision. **Subhadeep De:** Conceptualization, Methodology, Formal Analysis, Writing – review & editing, Supervision, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Instability caused by the cavity misalignment

In the method-4 as discussed in Section 2.3.3, the time-dependent displacement along all three directions are obtained from forced vibration simulation, for given input acceleration along each of the directions. Such angular and lateral displacements (misalignment) cause the optical power in the FM to redistribute into the HOM, of which the first-order mode is the most significant [55], however will introduce non-ignorable HOM. Considering a steady reflected power for PDH sideband light that is immediately reflected by the cavity mirror, the power of the interference signal between the PDH sidebands and the HOM is proportional to the HOM field amplitude. In case of the first HOM in Eq. (3), the interference signal due to lateral displacement $d_\circ$ is proportional to $d_\circ/\omega_\circ$, where $\omega_\circ$ is radius of the laser beam waist.

For symbolical distinction, let us consider X, Y, Z in the subscripts representing the directions of applied (forced) vibrations and x, y, z in the superscript are the directions of displacements. The two ends of the cavity are labelled as 1 and 2 as shown in Fig. 4. Therefore d_X^{y1} represents the displacement of end 1 from its original position along the Y-direction when the vibration is forcefully applied along the X-direction, as an example, and similarly others. The mean displacement along X when subjected to an acceleration along X is given by

$$d_X^x = \frac{d_X^{x1} + d_X^{x2}}{2} \tag{A.1}$$

The mean displacement d_X^x is scaled by the ratio of the vibration amplitude encountered by the cavity in operation to the applied forced vibration amplitude. Here we have assumed a linear relationship between the magnitudes of the applied vibration and its resultant displacement. This is represented by scaled mean displacement in the same direction d_X^{x*}. The field amplitude introduced in the higher order modes due to lateral displacement $d_\circ$ when subjected to vibration in the X axis is proportional to,

$$U_X^d = \sqrt{\left(\frac{d_X^{x*}}{\omega_\circ}\right)^2 + \left(\frac{d_X^{y*}}{\omega_\circ}\right)^2}. \tag{A.2}$$

Similarly, considering the first HOM in Eq. (4), the interference signal due to angular displacement θ is proportional to θ/θ_D, where $\theta_D = \lambda/(\pi\omega_\circ)$ with λ being the wavelength of the light. Differential displacement of the mirrors $D_X^x = d_X^{x2} - d_X^{x1}$ can be similarly scaled to D_X^{x*}. Angular misalignment of the cavity in the X axis is given by,

$$\theta_X^x = \tan^{-1}\left(\frac{D_X^{x*}}{L}\right). \tag{A.3}$$

The field amplitude introduced in the HOMs due to angular misalignment of the cavity due to vibrations along the X direction is proportional to,

$$U_X^\theta = \sqrt{\left(\frac{\theta_X^{x*}}{\theta_D}\right)^2 + \left(\frac{\theta_X^{y*}}{\theta_D}\right)^2}. \tag{A.4}$$

The angular and lateral displacements along each of the axes is computed and then combined to give the fractional frequency instability due to the impact of the HOM field amplitude on the PDH stabilization

loop. For example, the HOM field amplitude due to vibration along the X direction is proportional to,

$$U_X = \sqrt{\left(U_X^d\right)^2 + \left(U_X^\theta\right)^2}, \tag{A.5}$$

and the resultant when vibrations are applied along three directions is proportional to,

$$U = \sqrt{U_X^2 + U_Y^2 + U_Z^2}. \tag{A.6}$$

This unwanted HOM field amplitude is added to the FM's field amplitude, which acts as noise for the PDH error signal. This induces a fractional frequency instability (due to angular and lateral displacements causing the misalignment) as follows

$$\left(\frac{\Delta\upsilon}{\upsilon}\right)_{misalignment} = \frac{U\Delta_\lambda}{\upsilon}, \tag{A.7}$$

where Δ_λ is the cavity linewidth in Hz. The displacements along the optical axis of the cavity are considered for the applied forced vibration along each direction. For example, the relative displacement along the Z (i.e., the optical axis of the cavity) due to vibration applied along the X is $\Delta L_X = d_X^{z2} - d_X^{z1}$. Similar to the lateral displacement, this displacement is scaled to ΔL_X^* to account for the vibration encountered by the cavity in operation. The combined relative displacement along the optical axis due to vibration is then obtained by

$$\Delta L = \sqrt{\left(\Delta L_X^*\right)^2 + \left(\Delta L_Y^*\right)^2 + \left(\Delta L_Z^*\right)^2}. \tag{A.8}$$

The fractional frequency instability due to length change along the optical axis is given by,

$$\left(\frac{\Delta\upsilon}{\upsilon}\right)_{\Delta L/L} = -\frac{\Delta L}{L}. \tag{A.9}$$

Finally, combining instability due to length change along the optical axis and the instability due to angular and lateral displacements (misalignment), total fractional frequency instability due to vibration can be written as,

$$\left(\frac{\Delta\upsilon}{\upsilon}\right)_{total} = \sqrt{\left(\frac{\Delta\upsilon}{\upsilon}\right)^2_{misalignment} + \left(\frac{\Delta\upsilon}{\upsilon}\right)^2_{\Delta L/L}}. \tag{A.10}$$

Appendix B. Instability from residual amplitude modulation (RAM)

Electronic hardware noise accounts for the noise from various components of the PDH system i.e. the source laser, actuator, servo, and the frequency discriminator (including photodetector). This noise term is dominated by the source laser and actuator noise up to 10 Hz Fourier frequency, by amplitude noise of the digital-to-analog converter used in the servo from 10 Hz to 1 kHz, and by the frequency discriminator above 1 kHz (Fig. 14a). The ADEV plot of electronic hardware noise stays below the Brownian noise level of the cavity, as shown in Fig. 14b. In order to counter the frequency instability caused due to residual amplitude modulation (RAM), our PDH setup will employ an electro-optic modulator with an angled facet with respect to the incoming light beam, enabling us to reduce RAM induced frequency instability to the 75 ppm level with respect to the cavity linewidth [73]. The corresponding frequency instability of the PDH system due to RAM has been calculated to be $1.71(3)\times10^{-19}$ and the corresponding frequency dependence is depicted in Fig. 14. In this estimation we consider PDH locking of a commercial laser, that is available with us, having ADEV of $\sim 1\times10^{-10}$ in 1 s and using digital servo electronics having ADEV of $\sim 3.3\times10^{-18}$ in 1 s that we developed indigenously for this application. This analysis reveals that RAM has no material impact on the total cavity instability as its contribution is an order of magnitude smaller than the other dominant instabilities.

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